

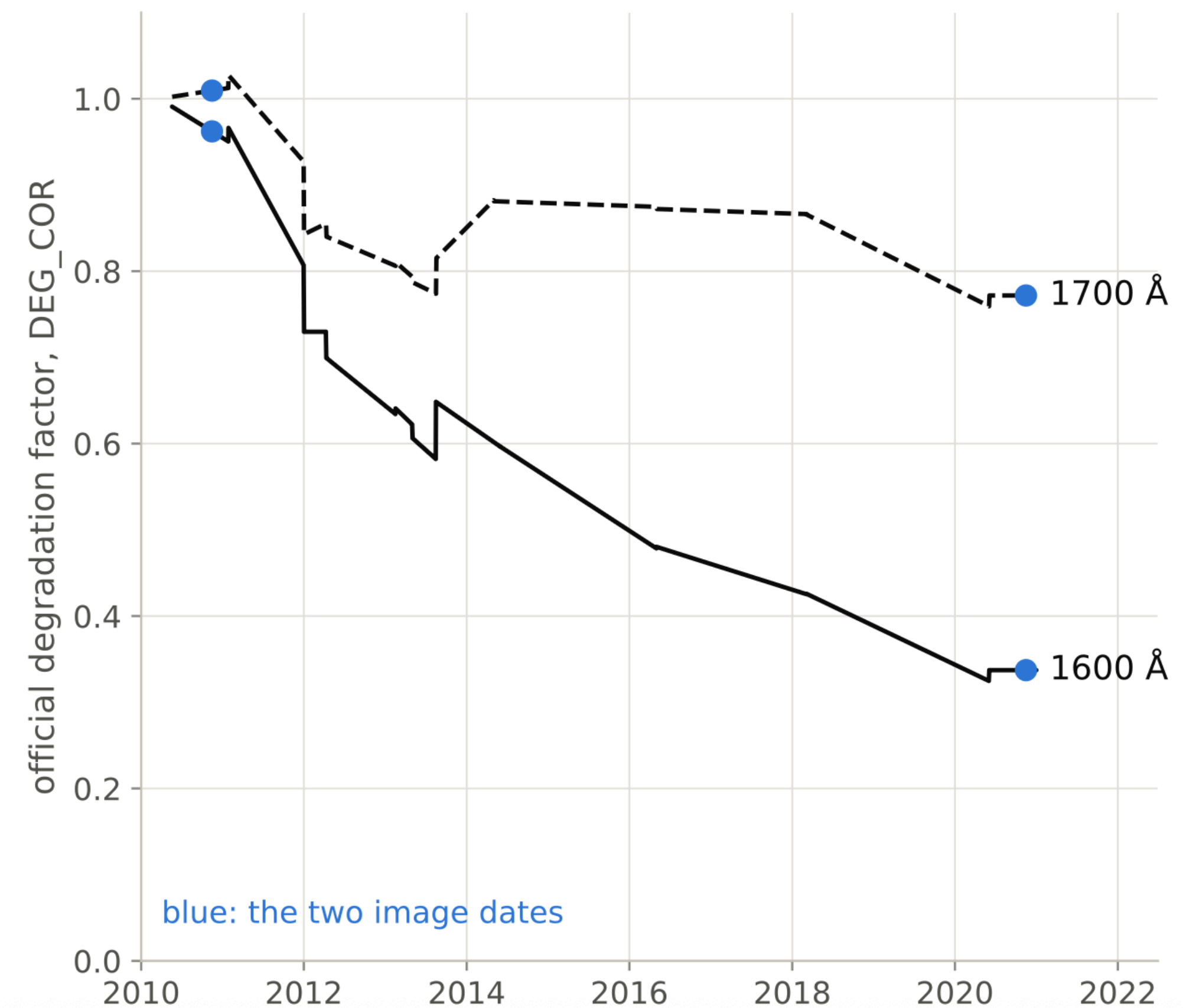
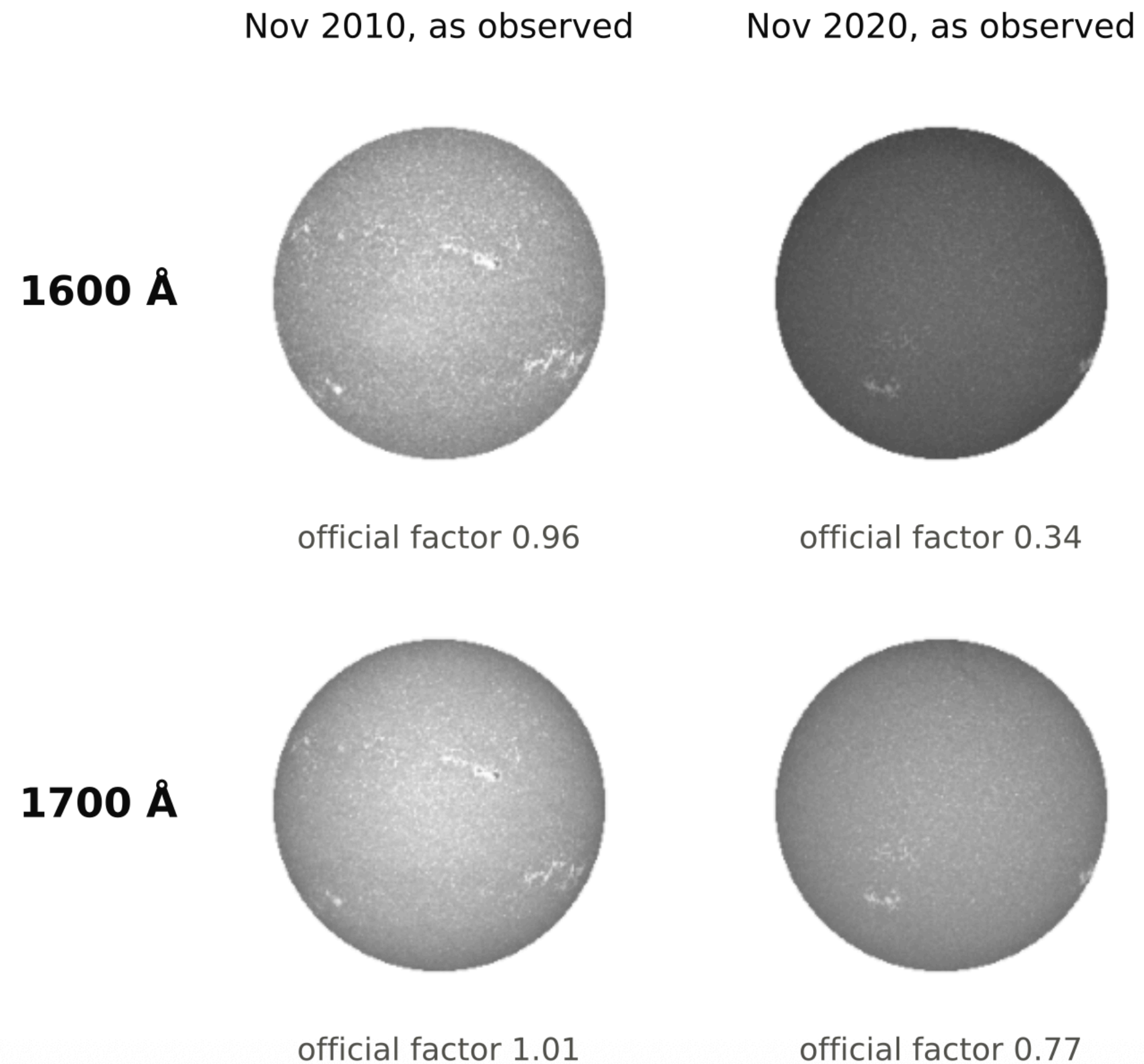
Auto-calibration for AIA UV images

Correct the time-dependent degradation

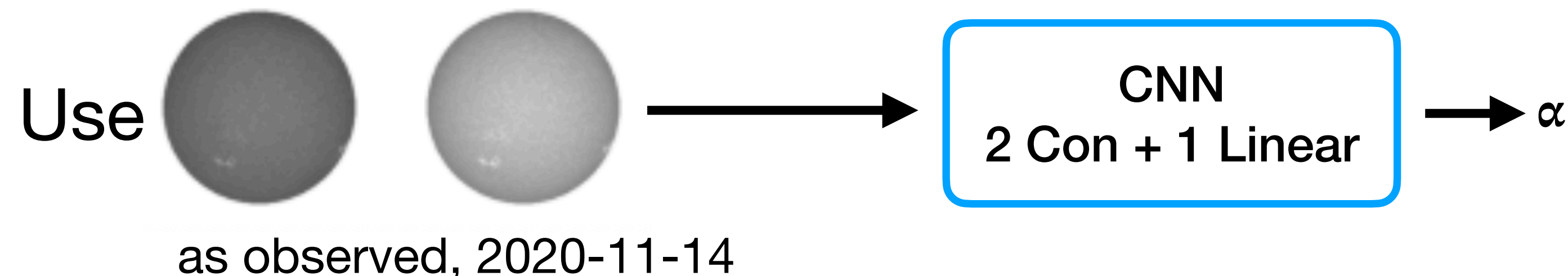
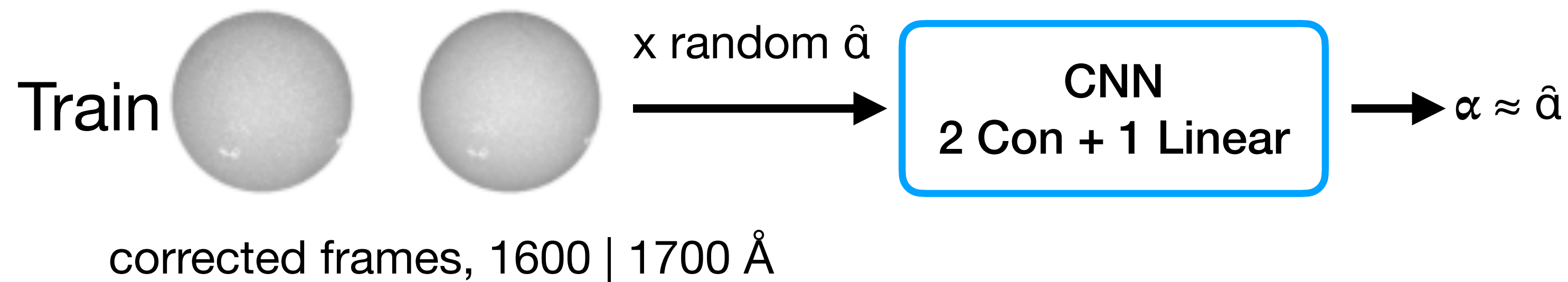
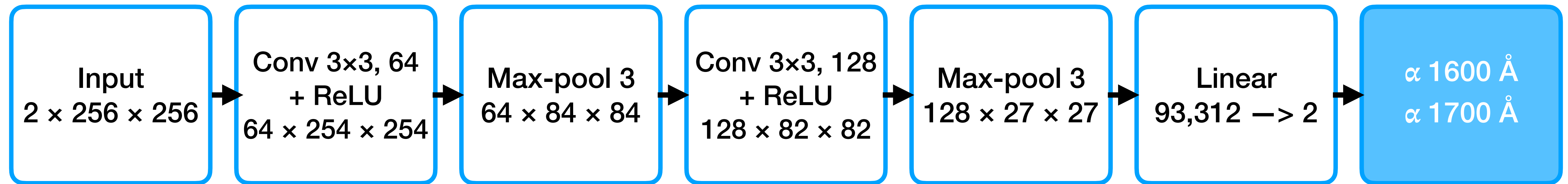
10/01/2026

Goal: auto-calibrate the dimming of AIA 1600 / 1700 Å images

- Data: SDOML-v2 (Galvez et al. 2019).
- Method: Dos Santos et al. 2021, A&A 648, A53.



A small CNN, trained on images dimmed by a known factor

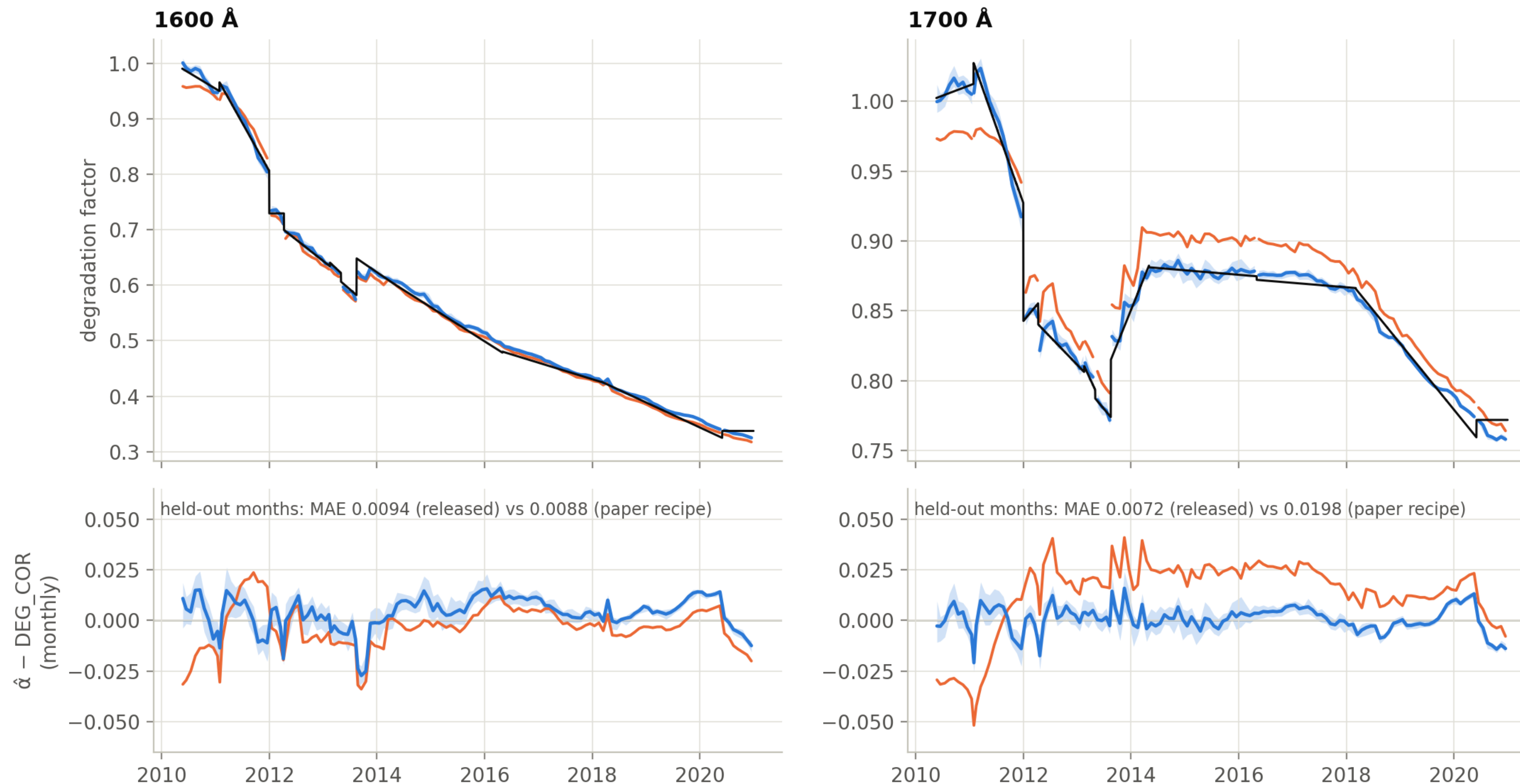


- Data: SDOML-v2 (Galvez et al. 2019): 3,826 daily pairs, 2010–2020.
- Train: January–July.
- Choose checkpoint: August – October.
- Held out: November–December.

Both follow the official factor; at 1700 Å the paper recipe drifts

Monthly medians over 2010–2020; band: 16th 84th percentile of single days (released model).

— official DEG_COR released model (band: 16th-84th percentile of days) paper recipe

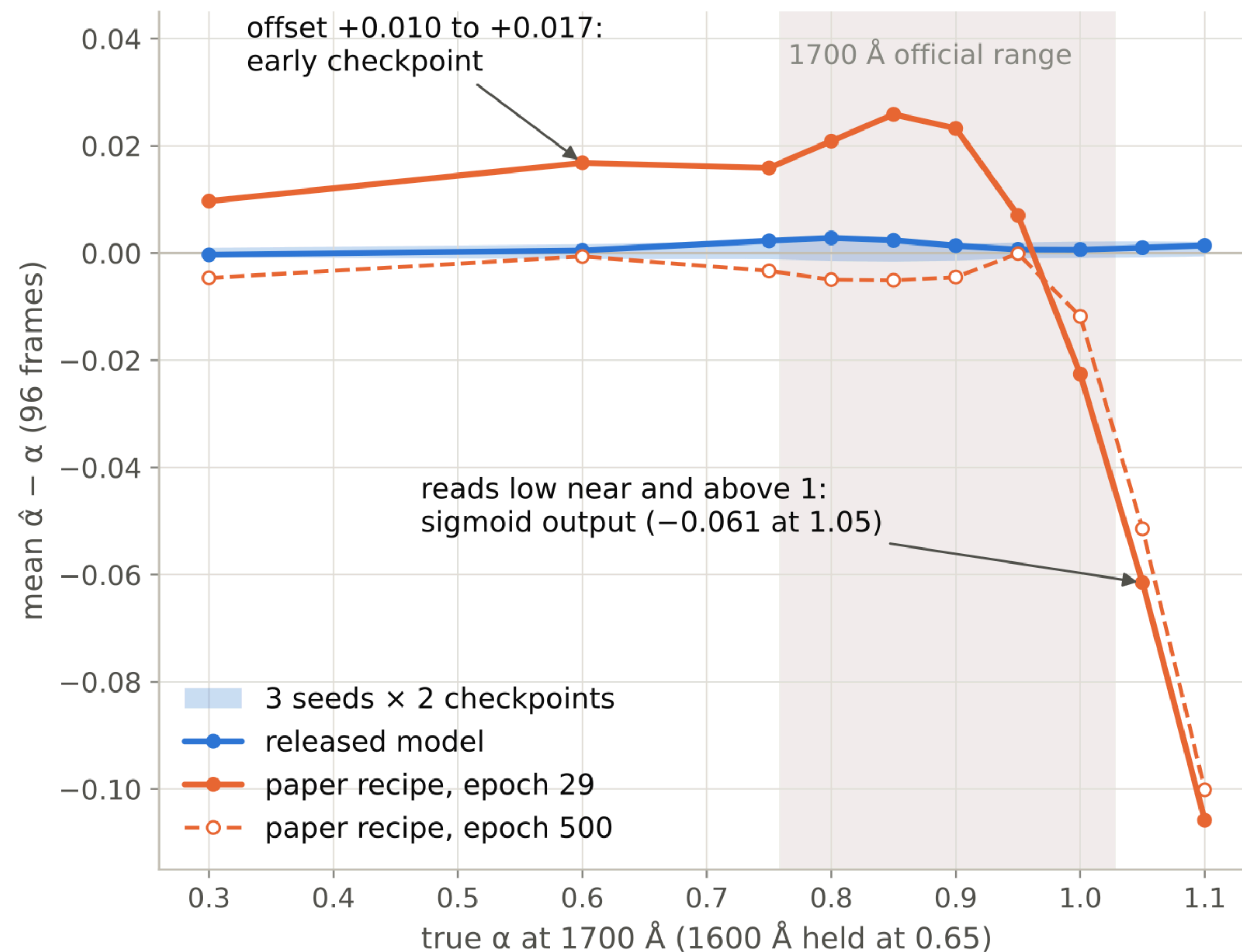


The official factor (DEG_COR) test checks recovery of a known dimming on real frames; it does not independently validate the official curve.

Why: an early checkpoint and a sigmoid output

Both bias the paper recipe at 1700 Å; the released recipe removes both.

1700



- **1 The checkpoint (larger effect)**

- cause: epoch 29 at a constant lr of 1e-3
- change: lr decay; pick by lowest test-month MSE, paper recipe's own epoch 500: error 0.0198.

- **2 The sigmoid output (structural)**

- cause: output 1, trained on 1
- change: linear head, trained up to = 1.25
- result: flat within 0.003 up to = 1.1

- **Outcome at 1700 Å**

- held-out error 0.0198 0.0072
- monthly RMS 0.022 0.006
- 1600 Å: 0.0088 0.0094, about the same